

1. Use the function $f(x) = -(x - 4)^2 - 3$ to answer problems 1a–1g.

a. Find the coordinates of the vertex for the parabola defined by $f(x)$.

b. Find the axis of symmetry of the parabola defined by $f(x)$.

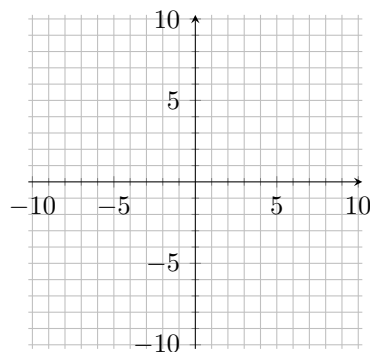
c. Find the maximum or minimum of $f(x)$.

d. Find the range of $f(x)$.

e. Find the domain of $f(x)$.

f. Find the y -intercept of the graph of $f(x)$.

g. Graph $f(x)$.



2. Use the function $g(x) = 4 + x^2 + 4x$ to answer problems 2a–2g.

a. Find the coordinates of the vertex for the parabola defined by $g(x)$.

b. Find the axis of symmetry of the parabola defined by $g(x)$.

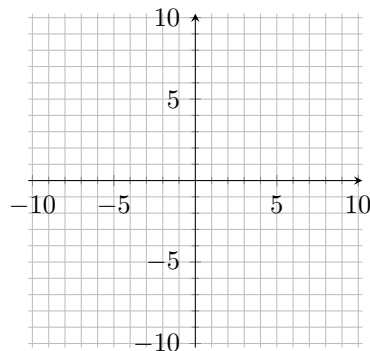
c. Find the range of $g(x)$.

d. Find the domain of $g(x)$.

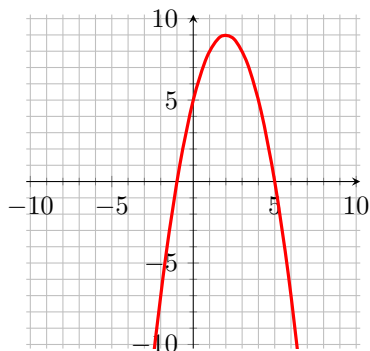
e. Find the x -intercepts (if any) of the graph of $g(x)$.

f. Find the y -intercept of the graph of $g(x)$.

g. Graph $g(x)$.



3. The graph of the function is given. Determine the function's equation.



(A) $x^2 - 4x$

(B) $x^2 + 4x + 5$

(C) $-x^2 - 4x + 5$

(D) $-x^2 + 4x + 5$

4. Determine if the following functions are polynomials.

a. $f(x) = 3 - \frac{3}{x^2}$

b. $f(x) = \pi x^3 - 2x^2 - 1$

c. $f(x) = \frac{2 - x^5}{3} + 2x$

d. $f(x) = x^{5/4} - x^2 + 9$

5. Find the degree of the following functions.

a. $f(x) = -2x + 6x^4 + 3x^2$

b. $f(x) = -12x + 2$

6. Use the leading coefficient test to determine the end behavior of the following functions.

a. $f(x) = 3x^2 + 2x - 3$

(A) falls to the left and falls to the right

(B) rises to the left and falls to the right

(C) rises to the left and rises to the right

(D) falls to the left and falls to the right

b. $f(x) = -7x^5 + 2x^2 - 3x + 100$

(A) falls to the left and falls to the right

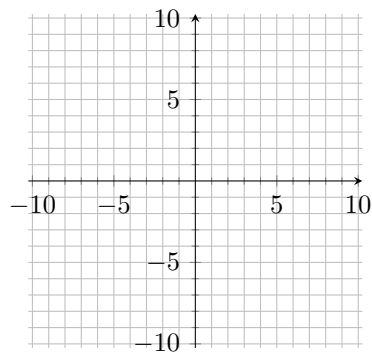
(B) rises to the left and falls to the right

(C) rises to the left and rises to the right

(D) falls to the left and falls to the right

7. Use the function $f(x) = 2x^2 - x^3$ to answer problems 7a–7b.

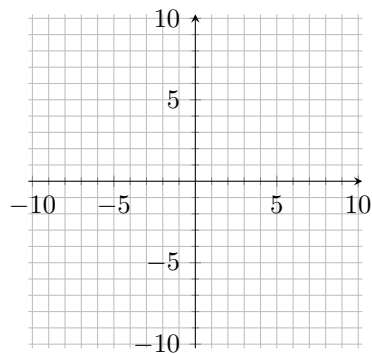
- a. Find when the x -intercepts of $f(x)$, and determine whether the graph crosses the x -axis, or touches the x -axis and turns around .



- b. Find the y -intercept of $f(x)$, and draw a rough sketch of $f(x)$ using your above answers.

8. Use the function $g(x) = x^4 - 4x^2$ to answer problems 8a–8b.

- a. Find when the x -intercepts of $g(x)$, and determine whether the graph crosses the x -axis, or touches the x -axis and turns around .



- b. Find the y -intercept of $g(x)$, and draw a rough sketch of $g(x)$ using your above answers.